PROBLEM 32

Particle with String on Spool

Original statement · Problems.pdf, p. 20

Rheonomic Spool Constraint

A heavy particle P of mass m is connected by a string of total length $l = 4\pi a$ to a cylindrical spool of radius a and center at O as shown in the figure. The spool rotates at a known constant angular rate ω counterclockwise. The string is massless, inelastic, thin, and remains always in tension during the motion of the particle. We define the inertial reference frame Oxy as shown in the figure. The other end of the string is attached to the spool at a point A of its perimeter, and we shall call B the instantaneous point of tangency between the straight segment of the string (the segment BP) and the spool. We will denote the angle between the local horizontal (axis Ox) and the radius OB as ϕ. Initially, $\mathbf{OA} = -a\mathbf{i}$, and B never reaches A.

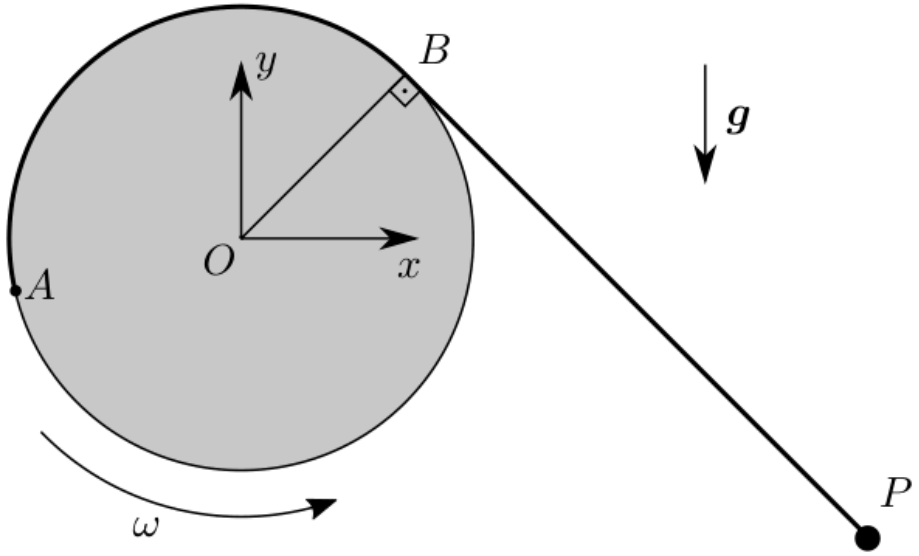


Figure: Spool of radius a centred at O rotating at ω, string wound from A to the tangency point B and straight segment BP (Problems.pdf, p. 20).

(a) Discuss the number of degrees of freedom of the motion of the particle.

(b) Write down the position, velocity and acceleration vectors of P with respect to Oxy as a function of the constants of the problem, time, the angle ϕ, and its derivatives.

(c) Write down the mathematical expression of the forces acting on P.

(d) Is the mechanical energy of P conserved? If so, write down the equation of conservation of mechanical energy; otherwise, explain why it is not conserved.

(e) Write down a closed set of equations that, if solved, would yield ϕ (and the other unknowns of the problem, if any) as a function of time. Note that you do not need to solve the equations.

(f) Consider now that $\omega = 0$. Can you write down a first integral of motion that only involves ϕ and $\dot{\phi}$ (but not $\ddot{\phi}$) in this case? You may want to rethink your previous answers for this particular case.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle P of mass m hangs from a string of length $l = 4\pi a$ that unwinds from a spool of radius a and centre O, which rotates counterclockwise at the constant rate ω. The string is massless, inextensible, thin and stays taut. One end is fixed to the spool at the point A of its perimeter; B is the point where the straight segment BP leaves the spool, and ϕ is the angle between Ox and OB. Initially $\overrightarrow{OA} = -a\,\mathbf{i}$ and B never reaches A.
- **Constraints:** P is a particle in the plane (2 coordinates). The string of fixed length that winds on the spool ties its position to the geometry: the free length BP is $s = l -$ (wrapped arc), and the attachment A turns with the spool, so the constraint depends explicitly on time (rheonomic, holonomic) when $\omega \neq 0$. It is unilateral ($T \geq 0$, taut string).
- **Degrees of freedom (part a):** $\mathrm{CDOF} = 2 - 1 = 1$, the angle ϕ.
- **Parameters:** $m, a, l = 4\pi a, \omega, g$.

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ with $\mathbf{j}$ vertical upward.
- **Basis at B:** $\mathbf{u}_r = \cos\phi\,\mathbf{i} + \sin\phi\,\mathbf{j}$ (radial) and $\mathbf{u}_t = -\sin\phi\,\mathbf{i} + \cos\phi\,\mathbf{j}$ (counterclockwise tangent), a rotation of $\{\mathbf{i}, \mathbf{j}\}$ by ϕ (matrix $\begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det = \cos^2\phi + \sin^2\phi = 1$; RR^T has diagonal entries $\cos^2\phi + \sin^2\phi = 1$ (and 1) and off-diagonal entries $\cos\phi\sin\phi - \sin\phi\cos\phi = 0$, so $RR^T = I$), with $\dot{\mathbf{u}}_r = \dot\phi\,\mathbf{u}_t$ and $\dot{\mathbf{u}}_t = -\dot\phi\,\mathbf{u}_r$. The straight segment leaves B clockwise, toward the lower right in the figure, i.e. along $-\mathbf{u}_t$.

Derivation

3.1 Kinematics

(a) Degrees of freedom

- The particle moves in the plane (2 coordinates) but the string constraint leaves exactly **one** degree of freedom, ϕ.

(b) Position, velocity and acceleration of P

Pedagogical Justification: the point B is at $a\,\mathbf{u}_r$ and P is at the distance s from B along $-\mathbf{u}_t$. The length s is the total length minus the wrapped arc. The attachment A is at the angle $\pi + \omega t$ (it turns with the spool), and the string is wound from B (angle ϕ) to A, so the wrapped arc is $a(\pi + \omega t - \phi)$:

$$s(t) = l - a\left(\pi + \omega t - \phi\right) = a\left(3\pi - \omega t + \phi\right), \qquad \dot s = a\left(\dot\phi - \omega\right)$$

$$\mathbf{r}_P = a\,\mathbf{u}_r - s\,\mathbf{u}_t$$

- **Velocity** (product rule and Poisson's formula):

$$\mathbf{v}_P = a\dot\phi\,\mathbf{u}_t - \dot s\,\mathbf{u}_t - s\left(-\dot\phi\,\mathbf{u}_r\right) = a\dot\phi\,\mathbf{u}_t - a\left(\dot\phi - \omega\right)\mathbf{u}_t + s\dot\phi\,\mathbf{u}_r = a\omega\,\mathbf{u}_t + s\dot\phi\,\mathbf{u}_r$$

- **Acceleration:**

$$\mathbf{a}_P = a\omega\left(-\dot{\phi}\,\mathbf{u}_r\right) + \dot{s}\dot{\phi}\,\mathbf{u}_r + s\ddot{\phi}\,\mathbf{u}_r + s\dot{\phi}\left(\dot{\phi}\,\mathbf{u}_t\right) = \left[s\ddot{\phi} + a\dot{\phi}^2 - 2a\omega\dot{\phi}\right]\mathbf{u}_r + s\dot{\phi}^2\,\mathbf{u}_t$$

(using $\dot{s}\dot{\phi} = a\left(\dot{\phi} - \omega\right)\dot{\phi}$).

3.2 Forces

(c) Forces on P

- **Weight:** $-mg\,\mathbf{j} = -mg\left(\sin\phi\,\mathbf{u}_r + \cos\phi\,\mathbf{u}_t\right)$.
- **Tension:** along the straight segment, toward B: $\mathbf{T} = +T\,\mathbf{u}_t$ with $T > 0$ (since $\mathbf{r}_B - \mathbf{r}_P = s\,\mathbf{u}_t$).
- **Inertia forces:** none. There is no other force: the string is the only constraint.

3.3 Equations of Motion

(e) Closed set of equations

> **Pedagogical Justification:** Newton's second law in S_0 gives two scalar equations for the two unknowns $\phi(t)$ and $T(t)$. The radial projection contains no tension:

$$\mathbf{u}_r:\ m\left[s\ddot{\phi} + a\dot{\phi}^2 - 2a\omega\dot{\phi}\right] = -mg\sin\phi, \qquad \mathbf{u}_t:\ ms\dot{\phi}^2 = T - mg\cos\phi$$

$$\boxed{a\left[\left(3\pi - \omega t + \phi\right)\ddot{\phi} - 2\omega\dot{\phi} + \dot{\phi}^2\right] = -g\sin\phi}, \qquad \boxed{T = ma\left(3\pi - \omega t + \phi\right)\dot{\phi}^2 + mg\cos\phi}$$

The first is the equation of motion for $\phi(t)$; the tension follows from the second once $\phi(t)$ is known.

(d) Energy

> **Pedagogical Justification:** the power of the tension on P is $\mathbf{T}\cdot\mathbf{v}_P = T\,\mathbf{u}_t\cdot\left(a\omega\,\mathbf{u}_t + s\dot{\phi}\,\mathbf{u}_r\right) = Ta\omega$. For $\omega \neq 0$ it is not zero (the contact point B moves with the spool), so the tension does work on P and the mechanical energy of P is **not conserved**. For $\omega = 0$ the power is zero, the constraint is scleronomic and the energy is conserved.

3.4 Integration and Answers

(f) Case $\omega = 0$

> **Pedagogical Justification:** with $\omega = 0$ the tension does no work and gravity is conservative, so the mechanical energy is a first integral that involves only ϕ and $\dot\phi$ (it contains no $\ddot\phi$). From $\mathbf{r}_P = a\,\mathbf{u}_r - s\,\mathbf{u}_t$ with $s = a(3\pi + \phi)$, the height is $y = a\sin\phi - a(3\pi+\phi)\cos\phi$, and $v^2 = s^2\dot\phi^2$ (from 3.1 with $\omega = 0$):

$$\boxed{E = \tfrac{1}{2}m\left[a(3\pi+\phi)\right]^2\dot\phi^2 + mga\left[\sin\phi - (3\pi+\phi)\cos\phi\right] = \text{const}}$$

Interpretation & Checks

- **Involute pendulum:** the system behaves like a pendulum whose effective length $s = a(3\pi - \omega t + \phi)$ changes as the string unwinds.
- **Dimensional check:** $[a\ddot\phi\, s/a] = \text{m s}^{-2}$, like $[g\sin\phi]$; $[T] = \left[ma\,(3\pi+\phi)\,\dot\phi^2\right] = \text{N}$.
- **Check against the official key:** the key of `Problems.pdf` gives the two equations $a\left[(3\pi - \omega t + \phi)\,\ddot\phi - 2\omega\dot\phi + \dot\phi^2\right] = -g\sin\phi$ and $ma\,(3\pi - \omega t + \phi)\,\dot\phi^2 = -mg\cos\phi + T$, the conservation of the mechanical energy for $\omega = 0$ and its quadrature $\frac{1}{2}m\left[a(3\pi+\phi)\right]^2\dot\phi^2 + mga\left[\sin\phi - (3\pi+\phi)\cos\phi\right] = E$. All coincide with the results above. Sympy reproduces $\mathbf{v}_P$ and $\mathbf{a}_P$ by differentiating $\mathbf{r}_P$, and numerical integration of the equation with $\omega = 0$ conserves the energy to 6×10^{-12} with $T > 0$.

Source: Problems.pdf